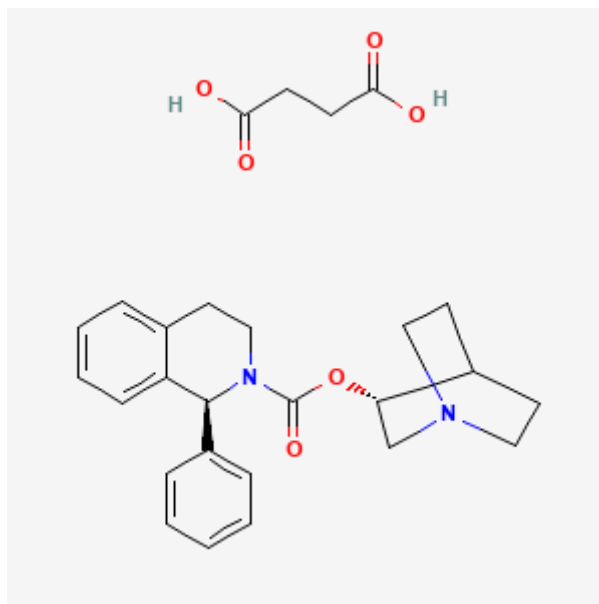


Leben Life Sciences Pvt. Ltd.

Product Name: Solifenacin Succinate Tablets BP 5 mg Film coated tablets 5 mg

P 2.2.1 ACTIVE INGREDIENT (DRUG SUBSTANCE)**a. General Information**

The active pharmaceutical ingredient is Solifenacin Succinate. It is a white to off-white crystalline powder. The chemical name is butanedioic acid, compounded with (1S)-1-phenyl-1,2,3,4-tetrahydroisoquinoline-2-carboxylic acid (1S,3R)-3-hydroxy-1-phenylpropyl ester, (1S)-1-phenyl-1,2,3,4-tetrahydroisoquinoline-2-carboxylic acid (1S,3R)-3-hydroxy-1-phenylpropyl ester. Its molecular formula is $C_{23}H_{32}N_2O_3 \cdot C_4H_6O_4$ and the molecular weight is 502.6 g/mol.

Structure**General Information**

Chemical Name	butanedioic acid, compounded with (1S)-1-phenyl-1,2,3,4-tetrahydroisoquinoline-2-carboxylic acid (1S,3R)-3-hydroxy-1-phenylpropyl ester, (1S)-1-phenyl-1,2,3,4-tetrahydroisoquinoline-2-carboxylic acid (1S,3R)-3-hydroxy-1-phenylpropyl ester
Molecular Formula	$C_{23}H_{32}N_2O_3 \cdot C_4H_6O_4$
Molecular Weight	502.6 g/mol

Mode of Action

Solifenacin Succinate is an antimuscarinic agent. It acts by inhibiting muscarinic receptors, which are involved in bladder muscle contraction. This action helps to reduce the frequency and urgency of urination associated with overactive bladder.

Absorption

Following oral administration, Solifenacin Succinate is absorbed from the gastrointestinal tract. Peak plasma concentrations are typically reached within 1 to 2 hours after dosing. The bioavailability is approximately 55%.

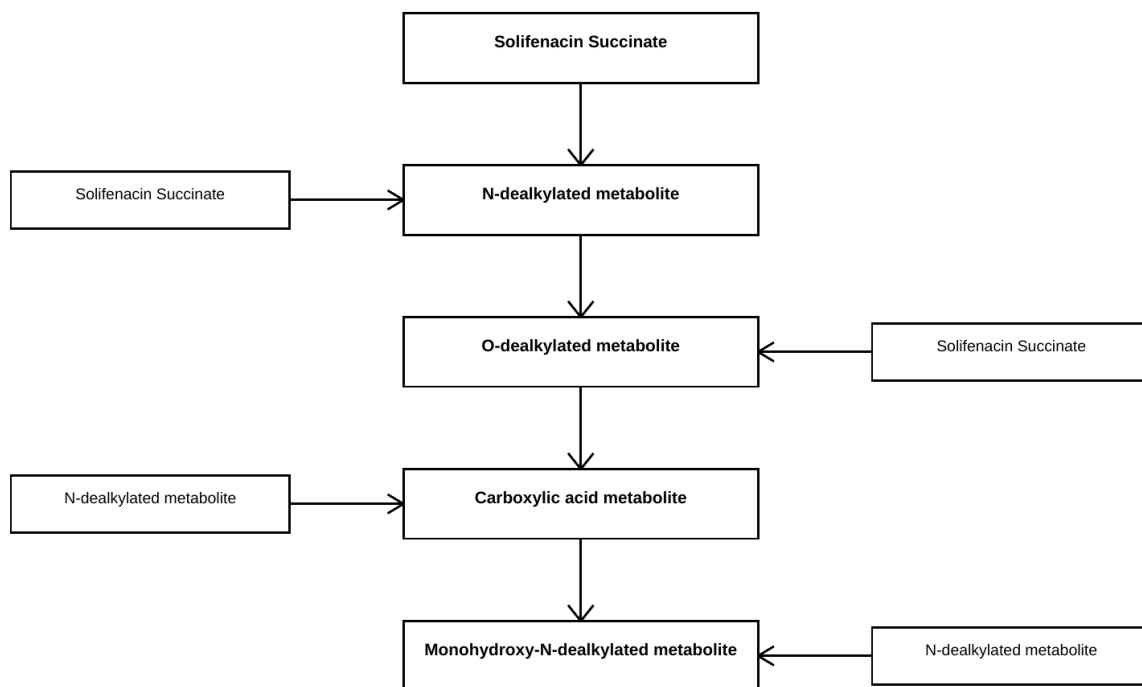
Distribution

Solifenacin Succinate is distributed throughout the body. It is approximately 98% bound to plasma proteins, primarily albumin and alpha-1-acid glycoprotein. The volume of distribution is approximately 113 liters.

Metabolism

Solifenacin Succinate undergoes extensive hepatic metabolism, primarily via the cytochrome P450 3A4 (CYP3A4) enzyme system. The main metabolic pathways include N-dealkylation and O-dealkylation, leading to the formation of several metabolites. The primary active metabolite is the N-dealkylated product.

Metabolism Flow Chart



Leben Life Sciences Pvt. Ltd.

Product Name: Solifenacin Succinate Tablets BP 5 mg Film coated tablets 5 mg

a. Excretion

Solifenacin Succinate is eliminated from the body through both renal and fecal routes. Approximately 70% of the administered dose is recovered in the urine, primarily as metabolites, and about 23% is recovered in the feces. The terminal elimination half-life is approximately 5-8 hours.

b. Physicochemical Properties

Solifenacin Succinate is a white to off-white crystalline powder. It is sparingly soluble in water and methanol, and slightly soluble in ethanol. The pH of a 1% aqueous solution is typically between 4.0 and 5.5. The melting point is approximately 130-135 °C. It exhibits polymorphism, with different crystalline forms potentially affecting its dissolution and bioavailability.

c. Preformulation Study and Selection of Excipients

Preformulation studies were conducted to understand the physicochemical properties of Solifenacin Succinate and to identify suitable excipients for the development of a stable and effective oral dosage form. These studies included characterization of solubility, stability, particle size, and compatibility with potential excipients. The goal was to select excipients that would ensure adequate drug release, tablet integrity, and long-term stability.

Preformulation Study Stages

The preformulation study involved several stages: 1. Characterization of the Active Pharmaceutical Ingredient (API): This included determining its physical form, solubility profile across a range of pH values, melting point, and hygroscopicity. 2. Excipient Screening: A range of common pharmaceutical excipients were screened for compatibility with Solifenacin Succinate using techniques such as Differential Scanning Calorimetry (DSC) and Fourier Transform Infrared Spectroscopy (FTIR). 3. Processability Assessment: Preliminary studies were performed to evaluate the API's flow properties and compressibility, essential for tablet manufacturing. 4. Stability Studies: The intrinsic stability of Solifenacin Succinate under various stress conditions (heat, humidity, light) was assessed to guide formulation and storage recommendations.

Table 1: Reference Product Specifications

SR. NO.	TEST PARAMETERS	SPECIFICATIONS	RLD (BATCH NO.)
1	Appearance	Light yellow coloured, round shaped, film coated tablet	
2	Identification	BP	
3	Assay	95.0% to 105.0%	
4	Dissolution	NLT 75% of labelled amount dissolved in 30 minutes	
5	Related Substances	As per BP	
6	Water Content	NMT 5.0%	
7	Uniformity of Dosage Units	As per BP	

d. Reference Product Information

The reference product is Solifenacin Succinate Tablets BP 5 mg, manufactured by Leben Life Sciences Pvt. Ltd. The product is a light yellow coloured, round shaped, film coated tablet containing 5 mg of Solifenacin Succinate. It is intended for oral administration and is packed in ALU/ALU blister packs of 10 tablets. The target product profile aims for a 36-month shelf life under controlled storage conditions, with specifications aligned to the British Pharmacopoeia.

Leben Life Sciences Pvt. Ltd.

Product Name: Solifenacin Succinate Tablets BP 5 mg Film coated tablets 5 mg

e. Excipients Information and Compatibility

The selected excipients for Solifenacin Succinate Tablets 5 mg are Lactose Monohydrate (diluent), Maize starch (diluent/binder), Hypromellose (binder), Magnesium Stearate (lubricant), and Readycoat Iron Oxide Yellow (film former). Compatibility studies, including DSC and FTIR, were performed to ensure no adverse interactions between Solifenacin Succinate and these excipients. The results indicated good compatibility, with no significant degradation or formation of new impurities observed under accelerated storage conditions. The chosen excipients are pharmacopoeially approved and suitable for direct compression or wet granulation processes, facilitating robust tablet manufacturing.

f. Excipient Selection Rationale

Lactose Monohydrate was selected as the primary diluent due to its excellent compressibility and inertness, providing bulk to the tablet. Maize starch serves a dual role as a diluent and binder, enhancing tablet hardness and aiding disintegration. Hypromellose was chosen as a binder to ensure adequate granule strength and tablet cohesion. Magnesium Stearate was incorporated as a lubricant to prevent sticking to the tablet tooling during compression. The film coating system, Readycoat Iron Oxide Yellow, was selected for its film-forming properties, providing a uniform coating, enhancing tablet appearance with a distinct color for identification, and offering protection to the tablet core. All excipients are of pharmacopoeial grade and have a well-established history of use in oral solid dosage forms.

Physicochemical Properties

Physicochemical properties are derived from the common_info and p2 sections of the technical intelligence.

Reference Product Information

Reference product information is synthesized from the common_info and p2 sections of the technical intelligence.

Reference Product Specification Note (optional)

Reference product specifications are extracted from the product_insights and formulation_development sections of the technical intelligence.

Excipients

Excipient information and rationale are synthesized from the common_info and p2 sections of the technical intelligence.